自觉遵守考场纪律　如考试作弊　此答卷无效

姓名_____

学号_____

密　封　线

东 南 大 学 考 试 试 卷（B 卷）

课程代码	BJSL0040	课程名称	离散数学	考试学期	2021-2022-3
适用专业	计算机类	考试形式	闭　卷	考试时长	120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

题目	一	二	三	四	五	六	七	八	总分
得分									
批阅人									

1. Choice questions (2 points for each question)

(1) Which of the following compound propositions is NOT a tautology? (　　)

A. $(\forall xF(x) \wedge G(y)) \rightarrow G(x)$

B. $\neg(\forall xF(x) \wedge \neg G(x)) \wedge (\neg\forall yF(y) \vee G(y))$

C. $(p\rightarrow q) \leftrightarrow (\neg p\vee q)$

D. $q \rightarrow (p\vee q)$

(2). The power set P(P(P(∅))) is equal to (　　)

A. $\{\{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}$.

B. $\{\varnothing, \{\varnothing, \{\varnothing\}\}, \{\varnothing\}\}$.

C. $\{\varnothing, \{\varnothing, \{\varnothing\}\}, \{\{\varnothing\}\}, \{\varnothing\}\}$.

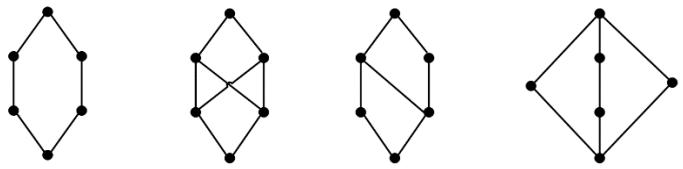
D. $\{\varnothing\}$.

(3). Given the set A={*a*, b, c, d}，and the relation R={<*a*, *a*>, <b, b>, <b, c>, <c, b>, <c, c>,<d, d>}, R does not satsify (　　).

A. Reflexive.　　B. Symmetric.

C. Transitive.　　D. Antisymmetric.

(4). Which one of the following posets is a distributed lattice？(　　)



A　　B　　C　　D

(5). For an undirected tree G with n ($n \geq 3$) vertices and m edges, which of the following

statements is wrong? ( )

A. G is connected with m=n-1.

B. Any edge of G is a bridge.

C. There exists a unique simple path between any two vertices of G.

D. G can have one leaf.

(6). G is a finite group G with an order |G|=8. Suppose *a* is an element of G. The order of *a* cannot be ( )

A. 1 B. 2 C.3 D. 4

(7). Let H and K be two subgroups of group (G, +). Which one of the following sets associated with operation + is also a subgroup of G? ( )

A. H∪K B. H∩K C. K-H D. H⊕K

(8). For which pair of (m, n) does the complete bipartite graph $K_{m,n}$ have a Hamilton circuit? ( )

A. (1, 1) B. (3, 4) C. (3, 3) D. (4, 3)

(9). G(V, E) is a simple undirected graph whose order is 7, the following four choices are the number of edges, which one ensures that G is connect? ( )

A. 15 B. 21 C.6 D. 16

(10). Suppose G is a simple planar graph, then δ(G) cannot be ( )

A. 3 B. 4 C. 5 D. 6

2. Completion (2 points for each question)

(1) Let P(*x*): *x* is a person; Q(*x*): *x* is left-hander; R(*x*): *x* drinks tea. Translating the following sentences by using quantifiers and P(*x*), Q(*x*), R(*x*)

a.) Some people drink tea. _____

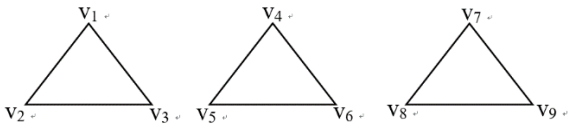
b.) Not all people are left-hander. _____

(2) Let A ={*a*, b, c}, R= ∅, write down all the properties that R satisfies_____

_____.

(3) Let G be a simple undirected graph with 7 nodes and 11 edges, then G at most have _____ connected components.

(4) The following graph G has three connected components. You should add at least _____ edges such that the new graph contains an Euler circuit.



(5) Let S={1, 2, 3, 5, 6, 10, 15, 30} and ⪯means exact division. Does <S, ⪯> form a Boolean lattice?_____(Fill "yes" or "no"). If you think <S, ⪯> is a Boolean lattice, then give out an isomorphic power set algebra _____ _____. If you think that <S, ⪯> is not a Boolean lattice, provide your reasons _____ _____.

(6) Let G be a simple connected planner graph with 10 vertices, then G at most have _____ edges.

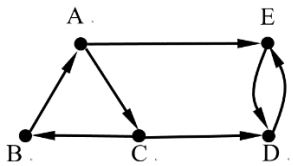
(7) <L, ∨, ∧, ', 0, 1> is a Boolean lattice, suppose $a \vee b = b$, then $a' \vee b =$_____.

(8) The node connectivity of the complete graph K_n is $\kappa(K_n) =$ _____.

3. Transform the following formula equivalently into a Prenex Normal Form: (10 points)

$\forall x \forall y(\exists z(P(x, z) \wedge P(y, z)) \rightarrow \exists u Q(x, y, u))$

4. **A directed graph G = (V, E) is given below: (10 points)**



(1) Please give out the adjacency matrix of G.

(2) How many paths whose length is equal to 3?

(3) How many paths from *A* to *D* whose length is less than or equal to 4?

(4) How many circuits from *D* to *D* whose length is less than or equal to 4?

5 Use proof by contradiction and necessary inference rules to prove that: (10 points)

$\{\forall x\,(P(x) \vee Q(x))\} \vDash \exists Q(x) \vee \forall x\, P(x)$

6 Suppose G is a simple planar graph with *n* (n≥3) nodes and *m* edges. Please prove that: (10 points)

(1) $m \leq 3n - 6$;

(2) If there does not exist a circuit with its length equal to 3, then $m \leq 2n - 4$.

7 Given a lattice <L, *, •>, if a, b, c$\in L$, $a * (b \bullet c) = (a * b) \bullet (a * c)$ holds. (10 points)

(1) Prove that $a \bullet (b * c) = (a \bullet b) * (a \bullet c)$ holds.

(2) Suppose <L, *, •> is also bounded, prove if $a \in L$ has its complement element b, then b is the unique complement element of a.

8 Let G be a group, and let H be a subgroup of G, please prove: (10 points)

(1) $Ha = Hb$ or $Ha \cap Hb = \varnothing$

(2) $\cup \{Ha \mid a \in G\} = G$